

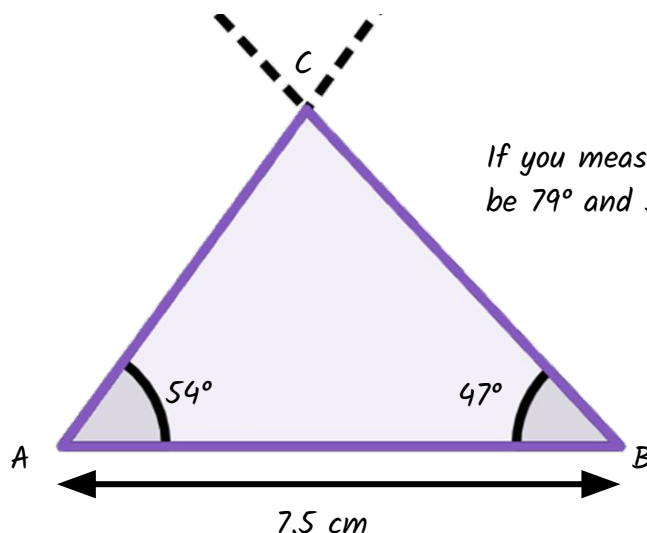


Congruent triangles (ASA and AAS)

Task A: Constructing triangles

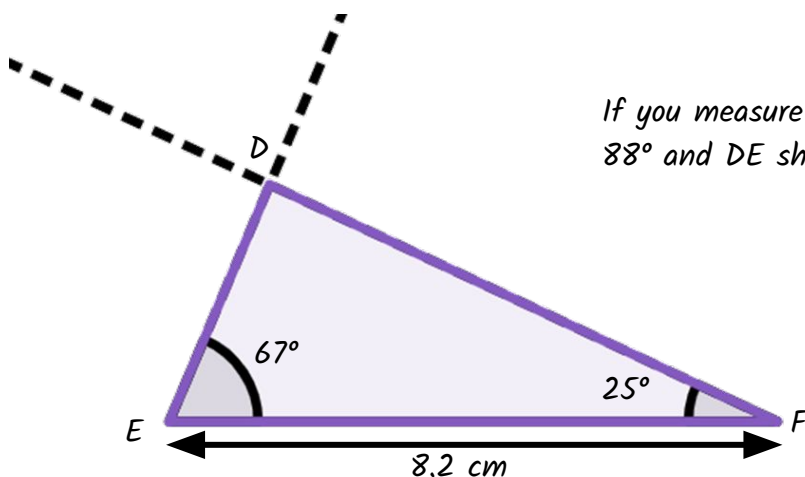
1) Construct the following triangles:

a) Triangle ABC, where $AB = 7.5$ cm, $\angle CAB = 54^\circ$ and $\angle ABC = 47^\circ$



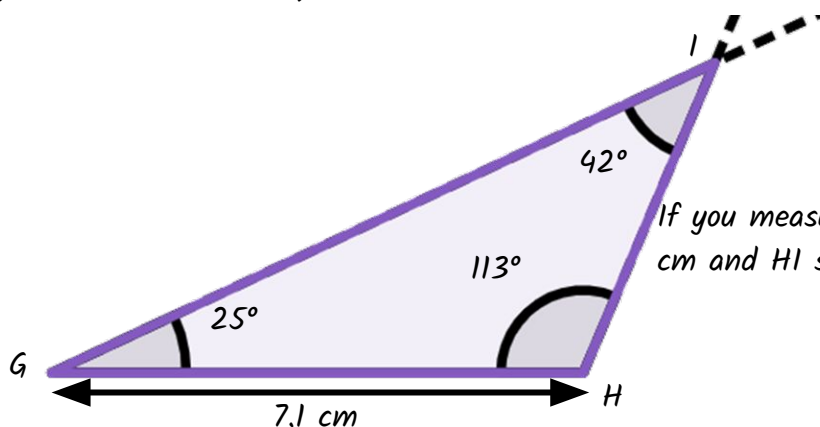
If you measure $\angle ACB$ it should be 79° and BC should be 6.2 cm

b) Triangle DEF, where $\angle DEF = 67^\circ$, $\angle EFD = 25^\circ$ and $EF = 8.2$ cm



If you measure $\angle EDF$ it should be 88° and DE should be 3.5 cm

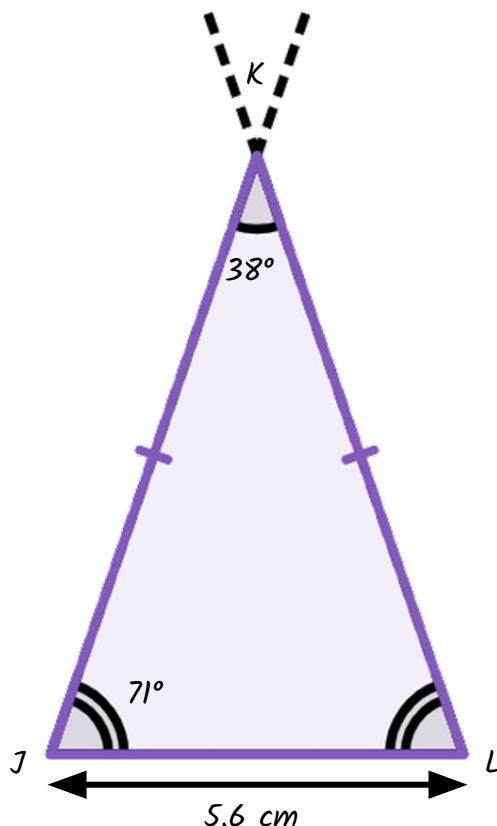
c) Triangle GHI, where $GH = 7.1$ cm, $\angle GHI = 113^\circ$ and $\angle HIG = 42^\circ$



If you measure GI it should be 9.8 cm and HI should be 4.5 cm



d) Triangle JKL, where $\angle JKL = 38^\circ$, $JL = 5.6$ cm and $KJ = KL$

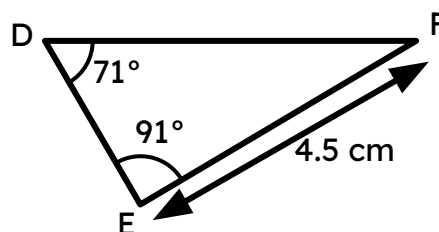
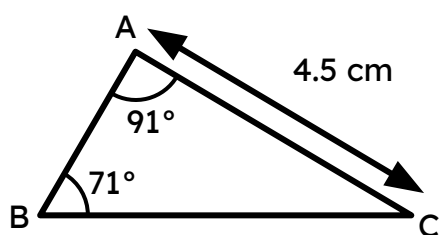


If you measure KJ and KL they should both be 8.6 cm

Task B: Justifying congruence by ASA/AAS

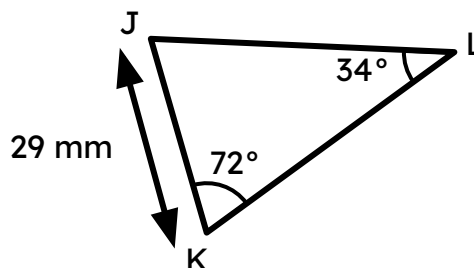
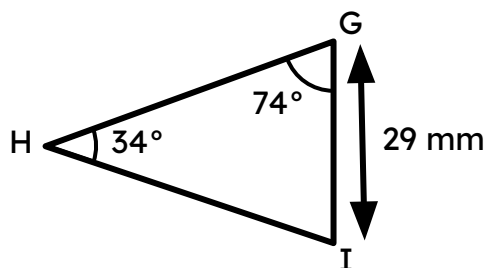
1) Prove each pair of triangles are congruent by ASA/AAS.

a)



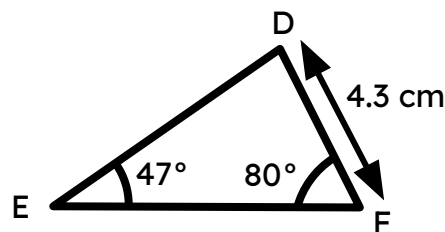
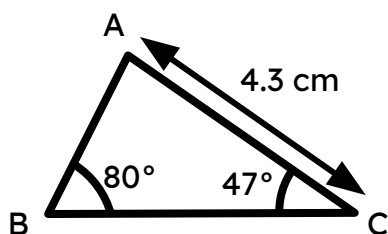
AC and EF are corresponding edges and equal, $\angle BAC = \angle DEF$ and $\angle ABC = \angle EDF$. Hence, congruent by AAS.

b)



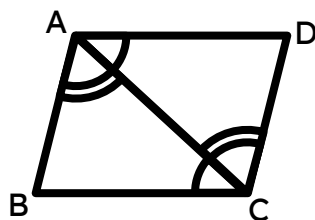
*$\angle GIH = 180 - (34 + 74) = 72^\circ$ and $\angle KJL = 180 - (34 + 72) = 74^\circ$
GI and JK are corresponding edges and equal, $\angle HGI = \angle KJL$ and $\angle GHI = \angle JKL$. Hence, congruent by ASA.*

2) Why are these two triangles not congruent by ASA?



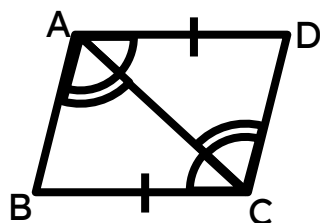
Although, the three interior angles of both triangles are the same.
The two edges of 4.3 cm are not corresponding edges in the triangles.

3) Given a parallelogram ABCD, prove that triangle ABC is congruent to triangle ACD by ASA/AAS.



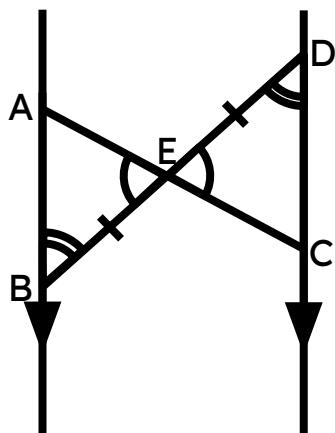
$\angle DAC = \angle ACB$ as they are equal interior alternate angles.
 AC is a shared side.
 $\angle DCA = \angle BAC$ as they are equal interior alternate angles.
Hence, triangle ABC is congruent to triangle ACD by ASA.

Alternatively:



$AD = BC$ as they are opposite sides of a parallelogram.
 $\angle DAC = \angle ACB$ as they are equal interior alternate angles.
 $\angle DCA = \angle BAC$ as they are equal interior alternate angles.
Hence, triangle ABC is congruent to triangle ACD by AAS.

4) Point E is the midpoint of AC and BD. Prove that triangle AEB is congruent to triangle ECD by ASA/AAS.



$DE = EB$, as E is the midpoint of BD.
 $\angle AEB = \angle DEC$ as vertically opposite angles.
 $\angle ABE = \angle EDC$, as they are equal interior alternate angles.
Hence, triangle AEB is congruent to triangle ECD by ASA.

